# THE COTANGENT COMPLEX OF THE ORIGIN OF THE AFFINE SPACE IS THE CONORMAL MODULE SHIFTED

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The statement is about a specific ring map and displays the conormal module; the author notes that other proofs exist but uses the fundamental triangle.

*Editorial note.* The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

*Allowed standard prerequisites (editorial).*

- The fundamental distinguished triangle for a composition of ring maps. equation-triangle, used at proof lines 2899-2908. Produces the triangle from which the cohomology is read.
- The cotangent complex of a polynomial algebra. Lemma lemma-cotangent-complex-polynomial-algebra, used at proof lines 2909-2911. Identifies the first term with the module of differentials.
- The vanishing of the cotangent complex of an isomorphism. Lemma lemma-when-zero, used at proof lines 2912-2913. Kills the middle term.
- The description of the cotangent complex of a surjection in terms of the conormal module. Lemma lemma-surjection, used at proof lines 2914-2915. Identifies the remaining cohomology group as I/I^2.

*Source reference.* Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 08QH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11060–11060. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 08RA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11088–11088. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 08R2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11080–11080. Navigation pointer into the Stacks

*Date*: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

**Lemma 0.2.** *Let $A = \mathbf{Z}[x_1, \ldots, x_n] \to B = \mathbf{Z}$ be the ring map which sends $x_i$ to $0$ for $i = 1, \ldots, n$. Let $I = (x_1, \ldots, x_n) \subset A$. Then $L_{B/A}$ is quasi-isomorphic to $I/I^2[1]$.*

*Proof.* There are several ways to prove this. For example one can explicitly construct a resolution of $B$ over $A$ and compute. We will use (08QS). Namely, consider the distinguished triangle

$$L_{\mathbf{Z}[x_1,\ldots,x_n]/\mathbf{Z}} \otimes_{\mathbf{Z}[x_1,\ldots,x_n]} \mathbf{Z} \to L_{\mathbf{Z}/\mathbf{Z}} \to L_{\mathbf{Z}/\mathbf{Z}[x_1,\ldots,x_n]} \to L_{\mathbf{Z}[x_1,\ldots,x_n]/\mathbf{Z}} \otimes_{\mathbf{Z}[x_1,\ldots,x_n]} \mathbf{Z}[1]$$

The complex $L_{\mathbf{Z}[x_1,\ldots,x_n]/\mathbf{Z}}$ is quasi-isomorphic to $\Omega_{\mathbf{Z}[x_1,\ldots,x_n]/\mathbf{Z}}$ by Lemma 08QH. The complex $L_{\mathbf{Z}/\mathbf{Z}}$ is zero in $D(\mathbf{Z})$ by Lemma 08R2. Thus we see that $L_{B/A}$ has only one nonzero cohomology group which is as described in the lemma by Lemma 08RA. □